The vertical distribution of juvenile salmon (*Oncorhynchus* spp.) and associated fishes in the Columbia River plume

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ABSTRACT

Simultaneous trawling at surface and at depth at one location off the Columbia River, Oregon, in June 2000 identified the depth distribution of juvenile salmonids and associated fishes. Juvenile salmon off the Columbia River were distributed primarily near the surface, within the upper 12 m. Highest densities of subyearling chinook salmon (*Oncorhynchus tshawytscha*) off the Columbia River were associated with high surface currents and decreasing tidal levels, with time of day possibly a co-factor. Densities of yearling chinook salmon increased with higher turbidity. Pacific herring (*Clupea pallasi*) was the most abundant and commonly caught forage fish, with density increasing at night, probably related to diel vertical migration. Catches of juvenile salmonids were not associated with catches of forage fishes. Daytime surface trawling appears to be an appropriate method for assessing the distribution and abundance of juvenile salmonids in marine habitats.

Key words: Columbia River, depth distribution, diel, Pacific salmon, surface trawling, tides

INTRODUCTION

Two of the key assumptions made when surveying marine populations are that the fishes or invertebrates of interest occupy the habitat that is being sampled, and that the survey gear is appropriate for that habitat (Gunderson, 1993). Nevertheless, marine fishes and invertebrates often show preferences for specific habitats depending on time of day, season, or other biophysical factors (Pitcher, 1993). For example, many fish species perform diel vertical migrations, which reduce the effectiveness of trawls fished at fixed depths (Blaxter and Holliday, 1963; Culley, 1971; Pillar and Barange, 1997). Unfortunately, fishery research programs often use sampling gear without investigating if it is appropriate for the identified habitat, or if diel behaviors by the fish could be affecting catches (Pillar and Barange, 1997).

There is widespread recognition that a significant portion of the ocean mortality of salmonids (*Oncorhynchus* spp.) occurs within the first few months at sea (Pearcy, 1992). Recent observed declines in the marine return rates of salmon in the Eastern North Pacific have led to the establishment of numerous research programs to identify the causes of marine mortality (see reviews by Beamish *et al.*, 2003; Brodeur *et al.*, 2003). The purpose of many field studies has been to examine the broad-scale distributions of juvenile salmon and relate them to the environmental conditions in which they are found. These studies require extensive surveys on predetermined grids or transects using purse seines or surface trawls. Each station is generally sampled only once, and sampling is almost always done during the daytime. Implicit in these sampling designs are the assumptions that diel variability in catch rate at a particular station is minimal and the target species (juvenile salmon) occupy the surface layer being sampled.

Concerns that either of these assumptions may not be valid led us to conduct a depth- and diel-stratified trawling experiment to examine vertical and temporal variability in juvenile salmon catches. To our knowledge, there have not been previous studies attempting to examine diel variability of salmon in catches at a single location using surface trawls. Pearcy and Fisher (1988) examined the diel catches of coho salmon (*O. kisutch*) at a single location based on four deployments of fine-mesh gill nets off the Oregon coast. They found the majority of juvenile coho salmon occurred in the upper four

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meters at all times of the day; however, their sampling only extended 12 m below the surface. Similar results were found for all species of salmonids collected in day and night gill net sets off Sitka, Alaska, although coho salmon were found deeper than the other four species of salmon [chum (*O. keta*), pink (*O. gorbuscha*), sockeye (*O. nerka*), and chinook salmon (*O. tshawytscha*); Jaenicke *et al.*, 1984]. Catches of immature salmon by troll gear set at different depths (Waddell *et al.*, 1992; Orsi and Wertheimer, 1995) revealed a broader distribution of depth of capture but, again, juvenile salmon were mainly collected in the upper 20 m of the water column. Beamish *et al.* (2000) summarized vertical catch data for juvenile coho salmon from 4 years of pelagic trawling in the protected waters of the Strait of Georgia. Consistent with previous results, they found the majority of juvenile coho salmon were in the upper 15 m, although some fish were collected deeper than 45 m. These collections were made between 06:00 and 18:00 h, did not include night samples, and were conducted over a broad area that may have introduced spatial variability.

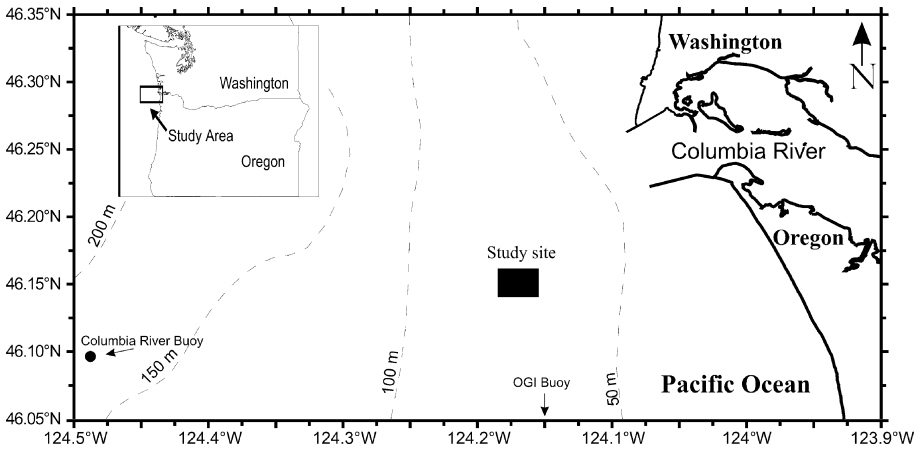
We report on the variability in physical oceanographic conditions and catch rates of juvenile salmon and other pelagic fish species in one location at two depth strata on the continental shelf within the Columbia River plume off Oregon. We conducted this study in the Columbia River plume because it is an area of high juvenile salmon abundance and to investigate possible relationships between it and salmonid abundance. We also wanted to determine if current sampling protocols used by us and other investigators for juvenile salmonids in marine waters are appropriate, given the variety of behaviors juvenile salmonids may demonstrate.

METHODS

Fish sampling

In June 2000, the 58-m fisheries research vessel CCGS *W.E. Ricker* conducted an initial surface trawl survey along an inshore–offshore transect just south of the mouth of the Columbia River, and identified a location which yielded high juvenile salmon catches (Fig. 1). The station, named CR-8 (46.15°N, 124.17°W), was approximately 13 km from shore and had a bottom depth of 60 m. At this station, the *Ricker* was joined by a 27-m commercial fishing vessel, the FV *Sea Eagle*, on June 21–22. Sampling was depth-stratified. Both vessels towed essentially identical trawl nets along parallel courses perpendicular to shore and approximately 0.7 km apart, with the *Sea Eagle* towing at the surface and the *Ricker* towing at mid-water depths (i.e. the *Ricker* headrope was below the fishing depth of the surface trawl of the *Sea Eagle*). Trawl speeds were 6.0 km h^{-1} (3.2 knots) and both vessels towed on a bearing of 150°. During trawl deployment, the *Sea Eagle* placed large floats on the trawl head rope (two on each wings and two in the center) and let out 137 m (75 fathoms) of warp. The *Ricker* did not place floats on the trawl headrope and let out approximately 183 m (100 fathoms) of warp to get the trawl to appropriate depth. The *Ricker* used a 264 Nordic rope trawl built by Nor'Eastern Trawl Systems, Inc.,[1] which has variable mesh sizes (162.6 cm at mouth to 8.9 cm at cod end), and a fishing mouth opening approximately 28 m wide × 12 m deep. The mouth opening and depth of the headrope were verified by the third-wire Simrad FS3300 backwards-looking net sounder. When fishing, this net was quickly lowered to depth (12–15 m

Figure 1. Location of study site off the mouth of the Columbia River occupied 21–22 June 2000.



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headrope depth), fished for half an hour, and then quickly retrieved to minimize surface contamination. Although the net fished the 12–24 m depth, it also fished for a short period of time in the surface layer during retrieval (approximately 5–10 min). The *Sea Eagle* used a pelagic rope trawl that had dimensions and meshes identical to the 264 Nordic rope trawl but was made in Astoria, OR. Both vessels used identical trawl doors (from Nor'Eastern Trawl systems) and 90.7-kg weight chains. After the first trawl set, trawling continued at regular intervals at the same location for 24 h, with the vessels alternating between the two starting points.

All fish captured in trawls were identified, counted, measured, and frozen for later laboratory analysis. If large catches of a fish species occurred during a trawl, a subsample of at least 50 was identified, counted, measured and individually weighed, and the rest weighed in bulk by species. Total numbers of that species captured during that trawl were later calculated by multiplying the number kg^{-1} (from the subsample) by the total kg weighed. Chinook (*O. tshawytscha*) and coho salmon were separated into age classes by length. We used salmon age nomenclature of Koo (1962), with number before the period indicating winters spent in freshwater after hatching and before migrating to sea, and the number after the period indicating winters spent in the ocean. Previous studies (Dawley *et al.*, 1985; Fisher and Pearcy, 1995) indicated that chinook salmon captured in June that measure <140 mm fork length (FL) are subyearlings (0.0-age – years in freshwater, years in ocean), 140–280 mm FL are yearlings (1.0-age), and >280 mm FL are 1-year ocean fish, either 0.1 or 1.1-age and older. All coho salmon less than 330 mm FL were considered 1.0 or yearling fish, and larger coho salmon were considered 1.1-age fish. Schabetsberger *et al.* (2003) reported on the feeding habits of juvenile salmon and zooplankton collected during this study.

Physical measurements

Vertical profiles of salinity, temperature, density and current measurements were collected before each trawl sample. On the CCGS *W.E. Ricker*, water-column profiling was done using a SeaBird Electronics SBE9+ conductivity, temperature and depth (CTD) instrument. A flow-through SBE 21 SeaCat thermosalinograph, with an intake near 3 m depth, provided continuous, underway sampling of temperature and salinity at 15-s intervals. Current measurements were collected with two RDI Acoustic Doppler Current Profilers (ADCP). The first, a 150-kHz, narrow-band ADCP, was mounted in a well in the *Ricker* hull. The second, a 300-kHz, broadband ADCP, was mounted on a buoy and deployed for 15 min every couple of hours to measure near-surface currents. Water transparency was measured with a transmissometer, and water samples for chlorophyll and nutrient analyses were collected from 3 m with a Niskin bottle. Aboard the FV *Sea Eagle*, CTD profiles were recorded with a Sea-Bird SBE 19 SeaCat profiler to within 5 m of the bottom. Predicted tidal heights during our sampling period were obtained for a tide station located at the mouth of the Columbia River. Swell height, swell period, wind speed and direction were obtained from the Columbia River Buoy (NOAA Buoy #46029, located at 46.12°N, 124.51°W) through NOAA's National Data Buoy Center website (http://www.ndbc.noaa.gov).

Statistical analysis

To permit comparisons of fish catches, all trawl data were standardized by transforming the raw catch data to densities (number/$10^6 m^3$). This was done by dividing the number of fish caught by the area of the trawl net mouth opening (336 m^2) as determined by the net sounder, times the over-ground distance fished by the trawl. Distance fished was calculated by computing the geographic distance between the beginning and ending trawl location using the geographic positioning system.

As salmon densities from the mid-water collections did not follow a normal distribution (chi-square goodness-of-fit test, $P < 0.05$), a non-parametric Mann–Whitney signed rank test was used to identify differences in median densities of salmonid catches (by species) between surface and mid-water samples. A *T*-test was used to identify any differences in mean length of salmonids between the various samples.

Transformed [$\log(n + 1)$] surface trawl juvenile salmon density data showed no difference from normality (X^2 all salmon species $P > 0.05$). Therefore, we used a general linear model (GLM) and associated analysis of variance (ANOVA) to investigate the relationship between physical variables and salmon catch densities in the surface layer. These physical variables included sea surface temperature, sea surface salinity, time after sunrise, tidal height at the mouth of the Columbia River, and their interactions.

Densities of small, short-lived pelagic fishes such as Pacific herring and smelt (i.e. forage fish) were not normally distributed (with or without transformation) (X^2 for all species $P < 0.05$). Differences in day versus night forage fish densities were identified by using a Mann–Whitney signed rank test.

RESULTS

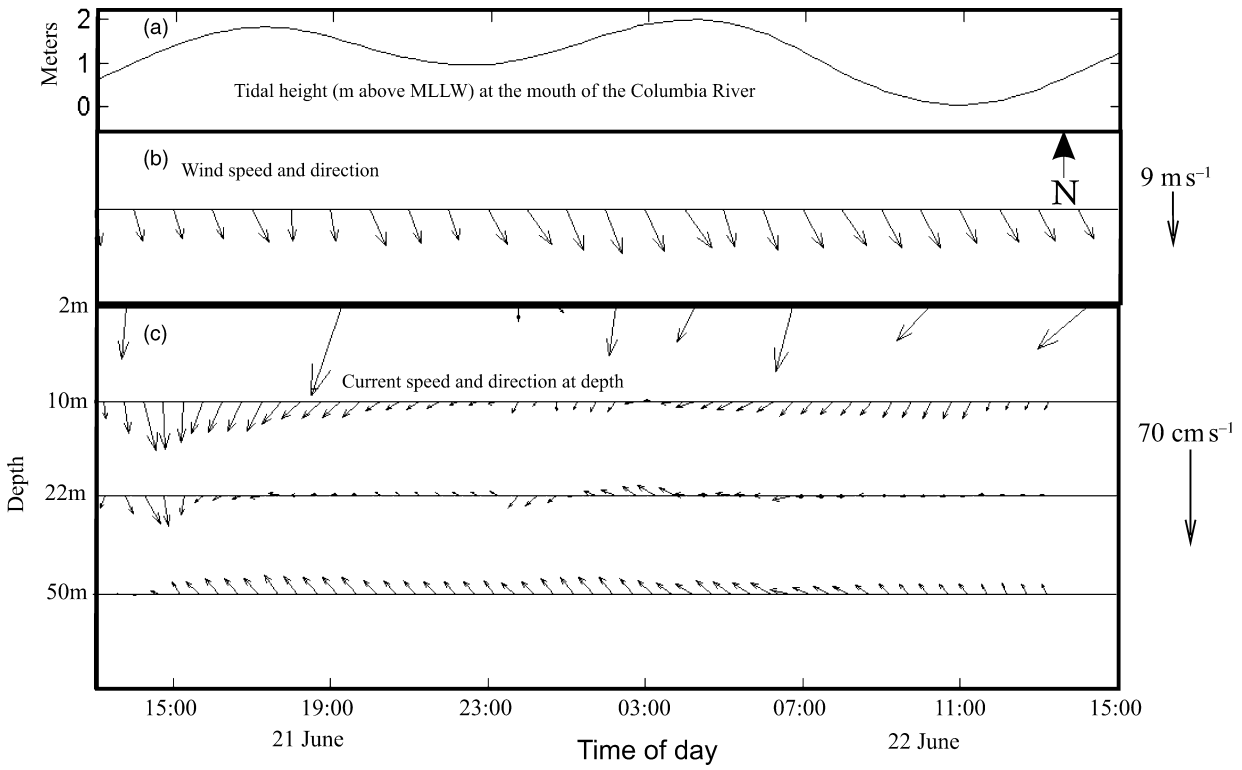
Physical variability

Winds during the sampling period were 3–9 m s^{-1} from the north–north-west or north-west (Fig. 2), and produced a typical summertime Columbia River plume orientation (Hickey, 1989). Significant wave height averaged 1.24 m with a period of 6.2 s. Satellite-derived advanced very high resolution radiometer (AVHRR) sea surface temperature showed the plume was advected to the south–south-west, but cloud cover limited AVHRR coverage. An offshore buoy south-west of the river mouth (OGI01; Fig. 1) corroborated the south-westward advection of plume waters, recording surface salinities from 20 to 28 PSU during the study period (A. Baptista, Oregon Graduate Institute, Beaverton, OR, USA, unpublished data).

Shipboard *in situ* data, instrumented buoys in the Columbia River estuary (http://www.ccalmr.ogi.edu/CORIE; A. Baptista, Oregon Graduate Institute, Beaverton, OR, USA, unpublished data), and AVHRR satellite data showed three surface-water masses and their respective salinities and temperatures: riverine plume (0.0 PSU, 17.5°C), deep shelf (33.5 PSU, 8.0°C), and surface shelf waters (28.0 PSU, 15.0°C). The surface shelf waters were dilute remnants of Columbia River plume waters that were advected north during a 5-day south-west wind event (June 10–14), then pushed south when north winds re-established.

Mean water-column depth was 63 m. Surface temperature and salinity ranged from 11.2 to 13.7°C and 19.4 to 31.0 PSU, respectively (Fig. 3). Surface waters were a mixture of riverine plume, deep-shelf, and surface-shelf water masses. Surface salinity and vertical stratification appeared to be responding to tidally pulsed water discharges from the Columbia River, identified by lower surface salinities and increasing surface current velocities during ebb tide. Surface currents were south-south-westward at 30–70 cm s^{-1} (Fig. 2), evidently in response to the forcing of the north-west wind and southward-flowing Columbia River plume. The CTD profiles revealed that the Columbia River plume waters were found primarily in the top 10–20 m of the water column (Fig. 3).

Figure 2. Tidal height (a), surface wind speed and direction (from Columbia River buoy) (b) and current velocities and direction at four different depths (c) off the mouth of the Columbia River, June 21–22, 2000. Arrows show the direction winds and currents are heading. Length of arrows indicates velocity. Legends showing the maximum velocity are shown to the right of the figures.



Water transmissivity showed a layer of turbid water, approximately 20-m deep, overlaying clear water (Fig. 3). Transmissivity was strongly related to salinities (correlation coefficient = 0.831) and thus a good indicator of Columbia River plume waters.

Trawl catches

We captured 17 different species of fish and eight invertebrate species (mostly gelatinous zooplankton) (Table 1). Pacific herring (*Clupea pallasi*) and whitebait smelt (*Allosmerus elongatus*) were the most abundant fish species captured. Salmon comprised 9% of the fish caught but were in every surface trawl (Fig. 4).

A total of 207 chinook salmon (*Oncorhynchus tshawytscha*) and 24 coho salmon (*O. kisutch*) were captured. Over 89% of the chinook salmon and 78% of the coho salmon were captured by the surface trawl, i.e. within 12 m of the surface (Fig. 4). Surprisingly, subyearling (0.0-age) chinook salmon, which are smaller and have a peak outmigration in summer (June/July) (Dawley *et al.*, 1986) after the yearling (1.0-age) chinook salmon peak outmigration in May, were the most abundant salmonid captured. Subyearling chinook salmon comprised over 47% of all salmon captured (Fig. 4). All coho salmon caught were 1-year ocean fish (1.1-age, identified by length).

Both salmon species were significantly more abundant in the surface 12 m than at mid-water depth (Mann–Whitney, $P < 0.05$; Table 2), indicating a preference for this habitat. The very small salmonid catches that did occur in the mid-water samples were probably related to contamination when the trawl net was retrieved through the surface (the net could not be opened or closed at the selected depth).

Salmonid densities varied widely. Highest densities (approximately $40/10^6 m^3$) were at the surface in the

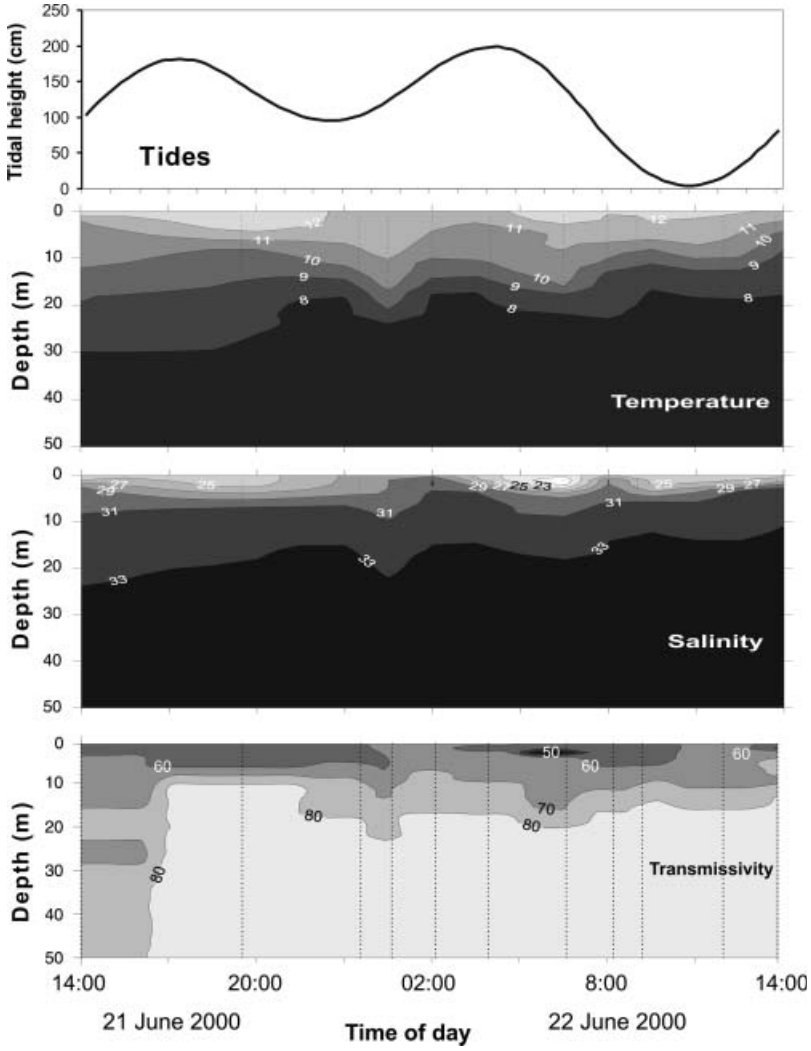


Figure 3. Vertical profiles of ocean temperature, salinity, and transmissivity over time at the study location, along with tidal height at the mouth of the Columbia River, June 21–22, 2000.

Table 1. Total number of individuals per species captured during depth stratified sampling off the mouth of the Columbia River.

Scientific name	Common name	Number ($N = 16$)
Cnidaria		
**Aequorea victoria*	Hydromedusa	1577
Scyphozoa	Jellyfish unidentified	50
**Chrysaora fuscescens*	Sea nettle	1992
**Aurelia labiata*	Moon jelly	18
Ctenophora		
**Pleurobrachia bachei*	Sea gooseberry	66
**Beroe* spp.	Combjelly	3
Mollusca		
Loligo opalescens	California market squid	2
Decapoda		
Crangon spp.	Sand shrimp	22
Urochordata		
**Thetys vagina*	Salp	78
Osteichthyes		
	Unidentified fish	3
Alosa sapidissima	American shad	3
Clupea pallasi	Pacific herring	1406
Engraulis mordax	Northern anchovy	80
Oncorhynchus kisutch	Coho salmon (age >1)	24
Oncorhynchus tshawytscha	Chinook salmon (age 0.0)	110
Oncorhynchus tshawytscha	Chinook salmon (age 1.0)	63
Oncorhynchus tshawytscha	Chinook salmon (age >1)	34
Osmeridae unidentified	Smelts unidentified	39
Thaleichthys pacificus	Eulachon	161
Allosmerus elongatus	Whitebait smelt	477
Microgadus proximus	Pacific tomcod	71
Sebastes melanops	Black rockfish	1
Ophiodon elongatus	Lingcod	3
Leptocottus armatus	Pacific staghorn sculpin	3
Liparis pulchellus	Showy snailfish	1
Anarrhichthys ocellatus	Wolf-eel	4
Citharichthys sordidus	Pacific sanddab	62
Platichthys stellatus	Starry flounder	2
Parophrys vetulus	English sole	1
Errex zachirus	Rex sole	9
Total number		6365

*Taxa that were noted to occur in the samples but not always counted.

morning, but time past dawn showed no significant relationship with any salmon catch (GLM ANOVA, $P > 0.05$). However, 0-age chinook salmon catches were related to tidal conditions, with highest salmonid catches related to low tide conditions (GLM ANOVA, $P = 0.0017$) (Fig. 4). High tide also occurred early in the morning.

Depth distributions of non-salmonid fish species also varied. Pacific herring, whitebait smelt, and northern anchovy (*Engraulis mordax*) were the dominant forage fish species in the surface collections, whereas eulachon (*Thaleichthys pacificus*) was the dominant forage fish species in subsurface waters (Fig. 5). Note that the highest catches of these non-salmonid fishes, mostly Pacific herring at the surface and eulachon in mid-water, occurred between the period of dusk and dawn, with overall highest catch at full darkness (2300). However, the small number of night collections and their strongly skewed catches limited statistical analysis. Catches of non-salmonid fishes did not appear to be related to tidal stage.

Sizes of salmonids

There was little variation in the size of chinook salmon captured over time (Fig. 6). Subyearling chinook salmon were easy to identify because of a length mode

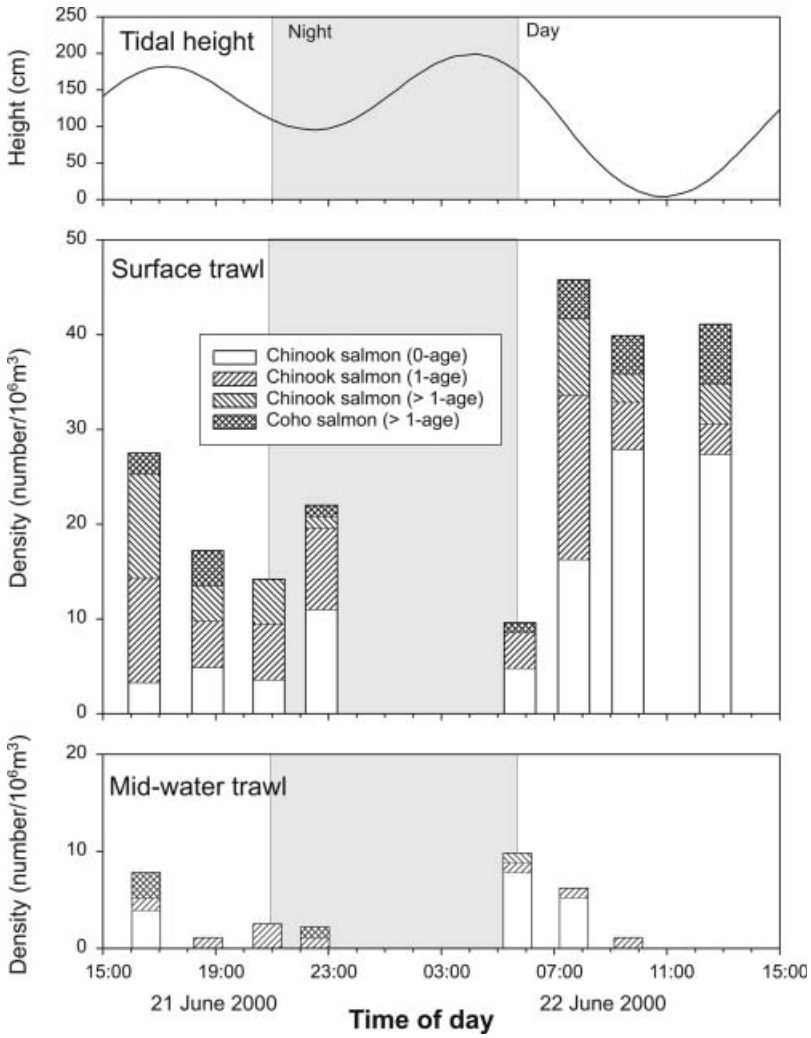


Figure 4. Densities of salmonids captured at the surface (0–12 m depth) and mid-water (12–18 m depth) by rope trawl off the mouth of the Columbia River, June 21–22, 2000. Tidal height at the mouth of the Columbia River and times of night and day are also shown.

Table 2. Median densities of salmonids captured off the Columbia River at different depths (surface and mid-depth) by rope trawl, during June 21–22, 2000. *P*-value is the difference between the medians using a Mann–Whitney test.

	Median density		
Species	Surface	Mid-depth	*P*
Chinook salmon (0.0-age)	7.94	0.00	0.0220
Chinook salmon (1.0-age)	5.46	1.11	0.0009
Chinook salmon (>1.0-age)	3.94	0.00	0.0025
Coho salmon (≥1.1-age)	2.93	0.00	0.0130

<140 mm FL (mean of 111 mm FL), while yearling chinook salmon were >150 mm FL (mean = 203 mm FL). Yearling chinook salmon (1.0-age) had a broader size range than subyearling chinook salmon (Fig. 6). No coho smolts were captured.

Catches relative to the physical environment

Subyearling (0.0-age) chinook salmon densities showed a relatively strong relationship to tidal height, with highest catches occurring at lowest tides when outgoing currents are strongest (GLM ANOVA, $P = 0.0079$; Table 3), indicating that currents emanating from the mouth of the Columbia River affected subyearling chinook distribution and abundance. Densities of yearling (1.0-age) chinook salmon showed a negative relationship with water transmissivity, with higher densities found at lower water clarity (GLM ANOVA, $P = 0.0616$) (Table 3).

The three most abundant forage fish species, particularly Pacific herring, showed a relationship to daylight, with highest catches occurring during dark-

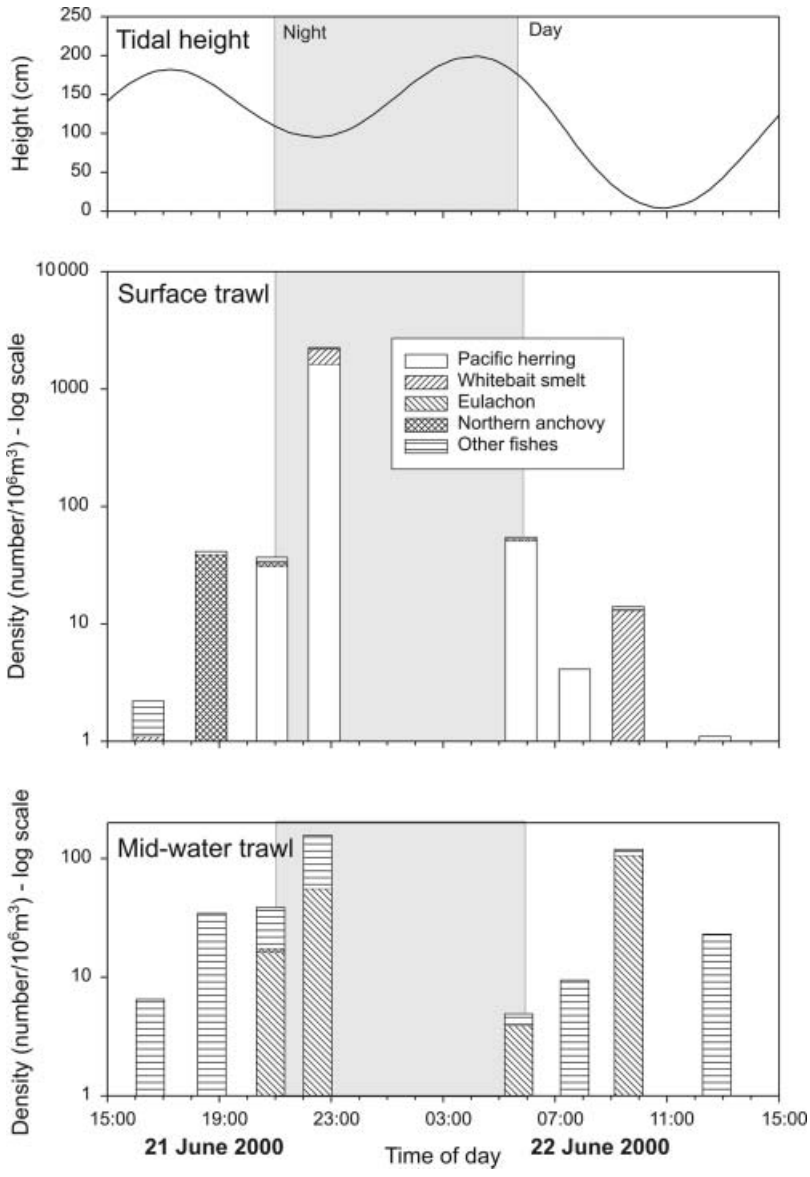


Figure 5. Densities of non-salmonid fishes captured at surface (0–12 m depth) and mid-water (12–18 m depth) by rope trawl off the mouth of the Columbia River, June 21–22, 2000. Tidal height at the mouth of the Columbia River and times of night and day are also shown.

ness. However we could not identify any significant differences between day/night catches (Mann–Whitney, $P > 0.05$) because of our limited sample size and the wide variation in catch densities.

The mixed semi-diurnal tides at the mouth of the Columbia River during this study were approaching a neap cycle and ranged over 2 m. Columbia River flow was 6100 $m^3 s^{-1}$, far below the spring freshet peak of 11 000 $m^3 s^{-1}$ that occurred in April (as measured at The Dalles, Oregon; US Geological Survey, unpublished data). The mean peak freshet flow from 1946–1999 is 11 500 $m^3 s^{-1}$, and typically occurs in June (Jay and Naik, 2002). The early timing of the 2000 freshet was, in part, a result of anthropogenic manipulation of river flow through dams, and reflects the trend in recent decades toward an earlier and decreasing peak Columbia River freshet due to hydropower operations.

DISCUSSION

Our data indicate that most juvenile chinook and coho salmon inhabit the water column from the surface to 12 m in the coastal oceanic environment. As such, characterizing the distribution and abundance patterns of juvenile chinook and coho salmon in the ocean using surface trawls is appropriate. We also found that the availability of juvenile salmon to sur-

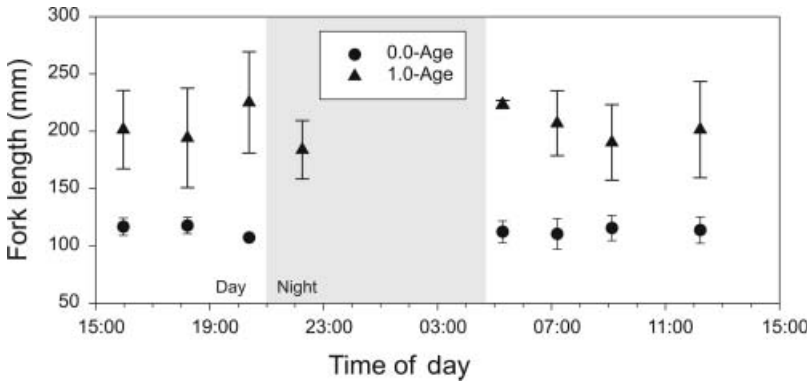


Figure 6. Mean lengths and standard errors of chinook salmon (*Oncorhynchus tshawytscha*) captured by surface rope trawl off the mouth of the Columbia River, June 21–22, 2000.

Table 3. Significant variables, coefficients, and analysis of variance *P*-values identified by general linear model that were significant predictors of juvenile salmon distribution and abundance.

	Salmon species/age class	
Variables	Chinook salmon (0.0-age)	Chinook salmon (1.0-age)
Constant	29.4407	11.3084
Time	ns	ns
Hours after sunrise	ns	ns
SST	ns	ns
SSS	ns	ns
Transmission	ns	−0.1614
Tide height	−0.1587	ns
Day/night	ns	ns
d.f.	7	6
R^2	82.67	44.26
P	0.0011	0.0616

No interaction terms were significant. SST, sea surface temperature; SSS, sea surface salinity; ns, non-significant.

face trawling in the Columbia River plume appears to be sensitive to time of day and/or tidal status. Juvenile salmon were more available to the trawls during the first part of the day (early morning to early afternoon). This period coincided with the peak ebb tide (i.e. tidal height was decreasing) during our study period. Tidal height appeared to be a particularly important variable determining the availability of subyearling chinook salmon off the Columbia River. Subyearling chinook salmon are relatively small (mean = 112 mm FL) and are probably easily swept out of the Columbia River estuary during periods of high current velocity. Surface current velocities at the mouth of the Columbia River can be higher than 3 m s^{-1} during peak ebb conditions (D. Jay, Oregon Graduate Institute, School of Science and Engineering, Beaverton, OR, personal communication). It is unlikely that a small salmon would be able to swim against these currents once entrained. Whether the 'flushing' of subyearling chinook salmon from the Columbia River estuary is a natural phenomenon is unknown. Subyearling chinook salmon typically reside in estuaries until summer or fall (Nicholas and Hankin, 1988).

The subsurface net, although catching significantly fewer salmonids than the surface net, may actually over-represent salmonid abundance because the net had to pass through the surface layer during setting and retrieval. While the subsurface trawl net does not fish effectively at the surface during deployment, it does during retrieval. During retrieval, the mouth of the net is open at the surface for approximately 5–10 min before being hauled on board. Without a net that opens and closes at depth, we have no way to assess the degree of extraneous sampling from the surface layer.

Beamish *et al.* (2000), trawling at four different depth strata in the Strait of Georgia, found a very large proportion (>50%) of all juvenile coho salmon were captured near the surface (0–15 m). Their catches at depth were probably also slightly contaminated by surface catches because they did not use an opening and closing net. Information on the depth distribution of juvenile chinook salmon were not reported.

Orsi and Wertheimer (1995), who collected salmon in Alaskan waters by trolling at depth, found that x.0-age coho salmon (1+ years in freshwater, no ocean winter) were more abundant near the surface (0.1–15 m). However, they found that x.0-age chinook salmon (0 + years in freshwater, no ocean winter) abundance peaked in 15.1–22.5 m depth strata. They also found older and larger chinook salmon were most abundant at depth (>22 m). Direct comparisons of these results with our data are not appropriate because the sizes of their salmonids were much larger than ours. For example, their smallest chinook salmon

(x.0-age) averaged 27.3 cm FL, while all our juvenile chinook salmon averaged <25 cm FL.

Although we found that most juvenile salmonids reside in the upper 12 m of the water column, it is possible that the location of our study site within the Columbia River plume influenced this finding. Undoubtedly many of the juvenile salmonids we captured were recent migrants from the Columbia River estuary and not actual 'ocean residents.' As such, these salmon may have been actively maintaining themselves in lower salinity surface-plume waters. Many of these fishes were probably hatchery-produced salmonids. Hatchery salmon smolts are known to have large amounts of body fat, which makes them relatively buoyant and may hinder their ability to migrate to depth (L. Weitkamp, personal communication, NOAA Fisheries, Seattle, WA, USA). Many of the recently migrated smolts may not have adjusted their physiology to the denser, high-salinity marine environment and thus were either actively or passively maintaining themselves near the surface in the Columbia River plume. While we found no relationship between sea-surface salinity and salmon catches, there was a relationship between 1.0-age chinook salmon densities and turbidity.

Subyearling chinook salmon, being relatively small, appear to have been swept offshore with the ebb tide and outgoing river flow. The relatively high surface velocities during peak ebb and low tide conditions and the corresponding peak in subyearling chinook catches support this conclusion. Past studies (Dawley *et al.*, 1985, 1986; Fisher and Pearcy, 1995) indicated that subyearling chinook salmon prefer shallow, nearshore ocean habitats in spring. We suspect that many, if not most, subyearling chinook salmon swim to shallow nearshore waters (i.e. just off the surf zone) after being swept out to sea by Columbia River flows. Purse seine catches in adjacent areas have indicated that very shallow nearshore habitats are important for small subyearling chinook salmon (<130 mm FL) (D. Miller, National Marine Fisheries Service, Hammond, OR, unpublished data). Unfortunately, our surface trawl gear cannot sample inshore of the 30-m isobath, since it initially sinks during deployment and can snag the bottom.

Catches of Pacific herring increased at night. Since the advent of sonar, the diel vertical behavior for clupeids has been very well documented (Blaxter and Holliday, 1963; Culley, 1971). During dusk, clupeid schools generally move toward the surface and break up, often forming dispersed schools near the surface at night. This movement appears to be related to both ambient light intensity and background light intensity (i.e. light levels below the schools) (Giannoulaki *et al.*, 1999). However, fishes may also be following prey resources. Schabetsberger *et al.* (2003) studying the zooplankton concurrently with our study, identified large increases in zooplankton densities at night, particularly copepods, hyperiid amphipods, and decapod larvae. Pacific herring and other forage fishes are known to feed at dusk and dawn on copepods and amphipods (Emmett *et al.*, 1991).

Daytime surface trawls appear to be an effective and appropriate method for sampling juvenile salmon in the ocean. However, our data for nighttime catches are limited. More surface trawling comparisons need to be conducted at night and day to confirm that juvenile salmon are equally available. Simultaneous stratified surface trawling also needs to be conducted to confirm that yearling juvenile coho salmon reside primarily in the top 12 m of the water column during the day and night. Furthermore, the use of daytime surface trawls to quantify abundance of clupeids, and perhaps large piscivorous fishes such as mackerel, is probably not valid. Surface trawling at night appears to be the most effective and appropriate method for sampling these species (Mais, 1974; Emmett *et al.*, 2001).

This study was located off the mouth of the Columbia River. While providing valuable information with regard to vertical distribution of salmonids and other species, our catches were affected by dynamic flow conditions from the Columbia River. This was identified by the significant relationship between tidal height and 0.0-age chinook salmon catches. It is entirely possible that in the environment off the Columbia River, salmonids are at the surface because of features associated with the plume (lower salinities, high turbidity, fronts, and eddies). It would be valuable to confirm that juvenile salmonids reside near the surface in other coastal regions and identify the bio/physical features with which they are associated.

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